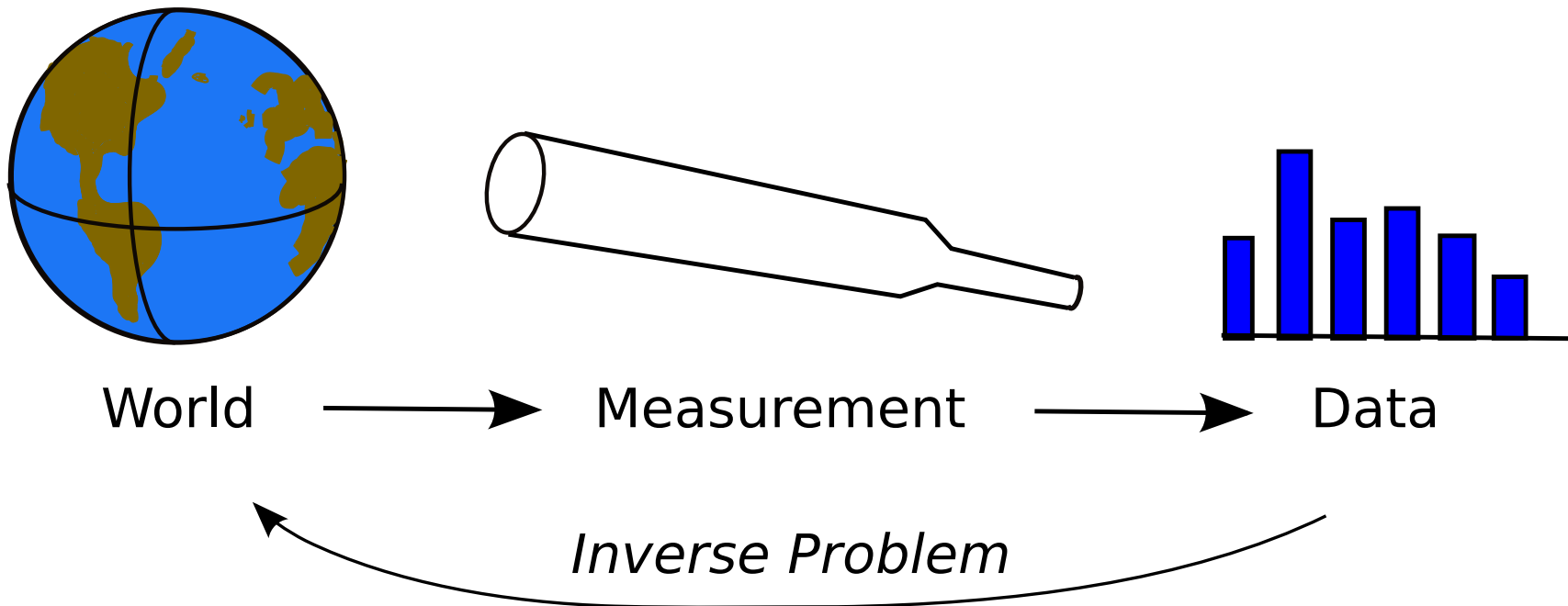


# Mathematics of Machine Learning

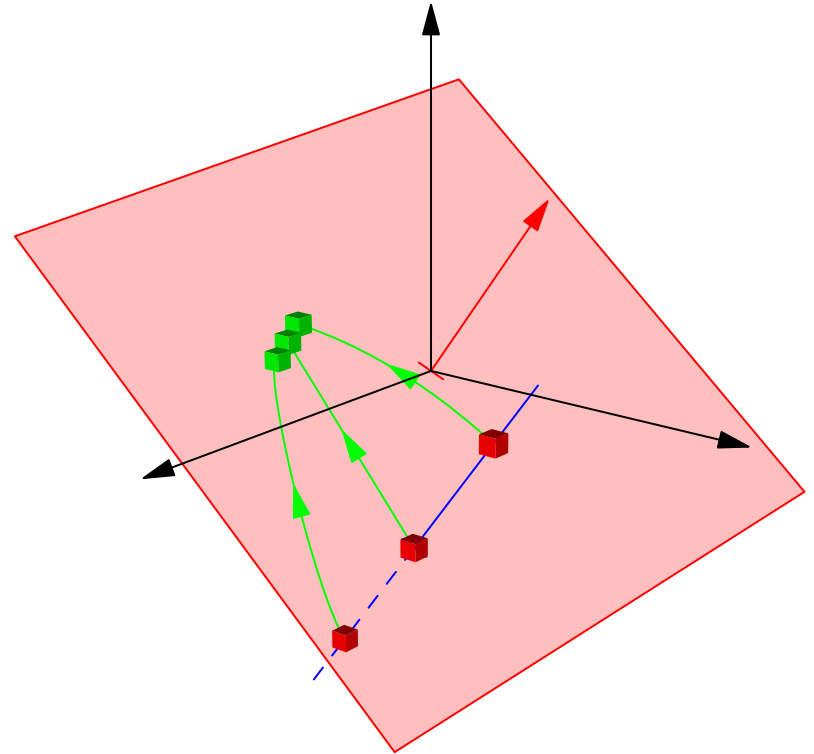
## *Understand Mappings*



*Mappings, Linear Maps, Solving Linear Systems*

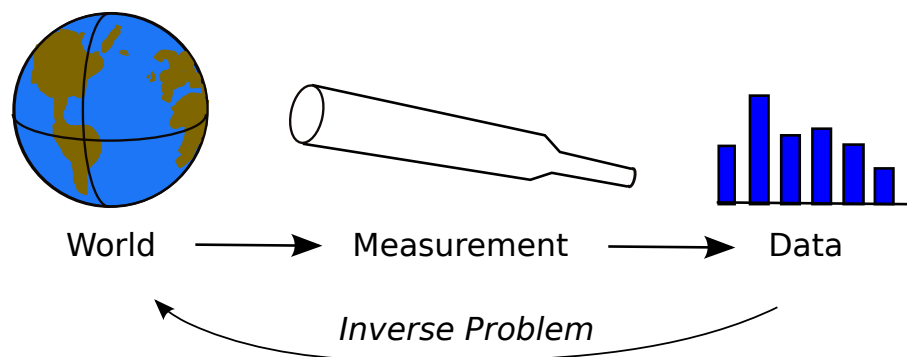
# Outline

1. **Mappings**
2. Linear Maps



# Transforming Data

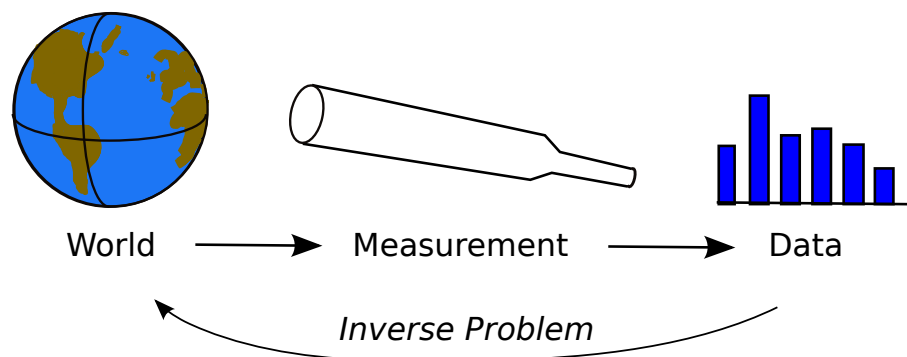
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- Although our mappings are not necessarily linear in either direction we learn a lot by understanding linear operators

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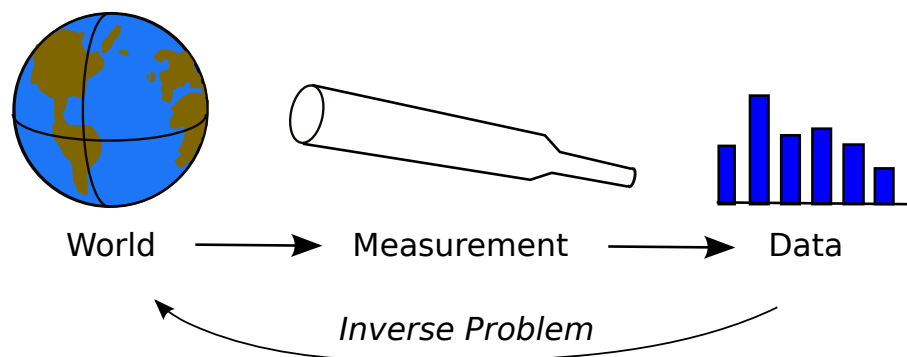
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# Inverse Problems

- Given  $m$  observations  $\{(\mathbf{x}_k, y_k) | k = 1, \dots, m\}$  and  $p$  unknown  $\mathbf{w} = (w_1, w_2, \dots, w_p)^\top$  such that  $\mathbf{x}_k^\top \mathbf{w} = y_k$  then to find  $\mathbf{w}$
- Define the *design matrix* as the matrix of feature vectors

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_m^\top \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mp} \end{pmatrix}$$

- and the target vector  $\mathbf{y} = (y_1, y_2, \dots, y_m)^\top$
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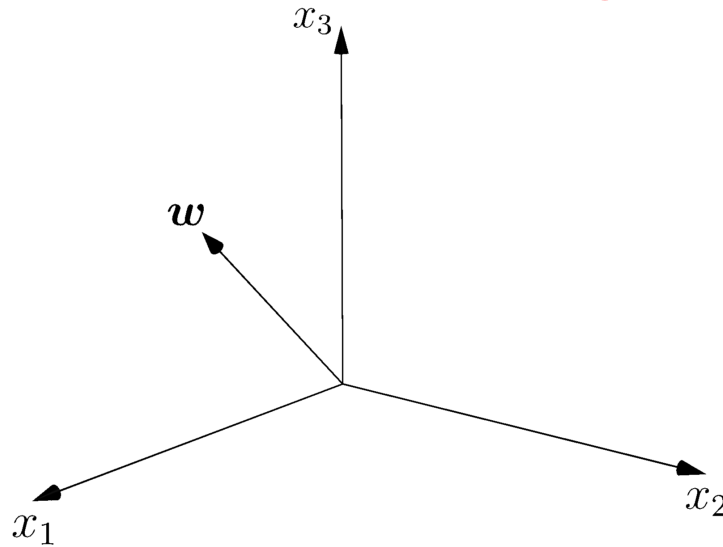
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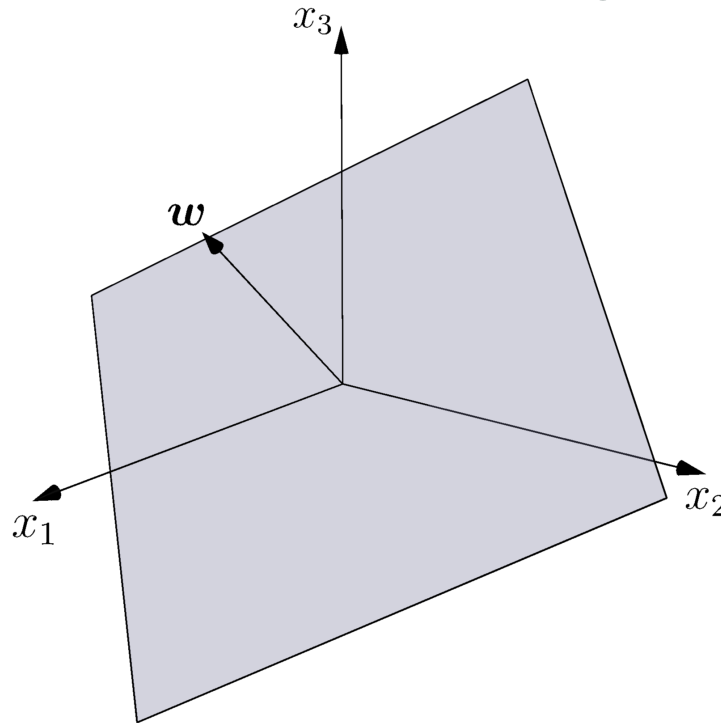
- $x_k^T w$  depends on distance from separating



- If  $m > p$  then  $X$  isn't square so doesn't have an inverse
- Worse unless the data is accurate  $y \approx Xw \Rightarrow$  no “solution”
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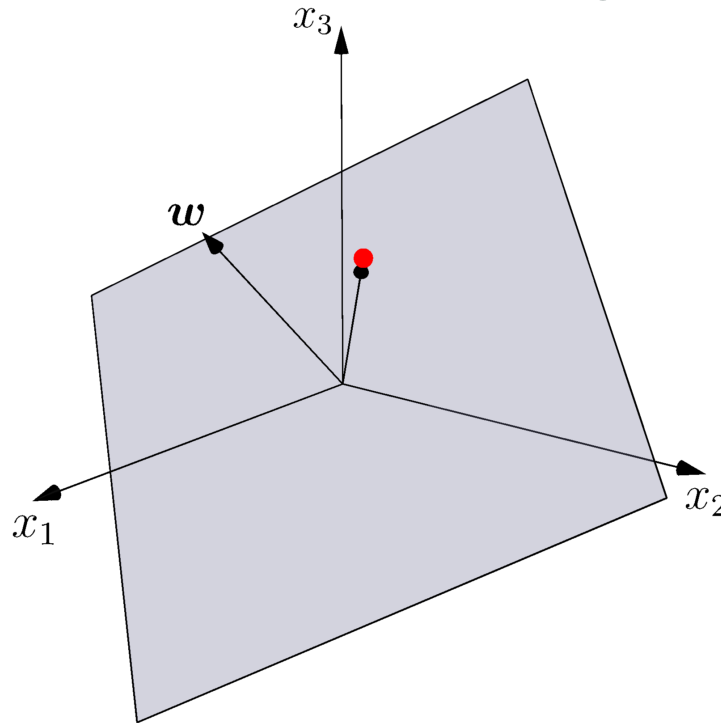
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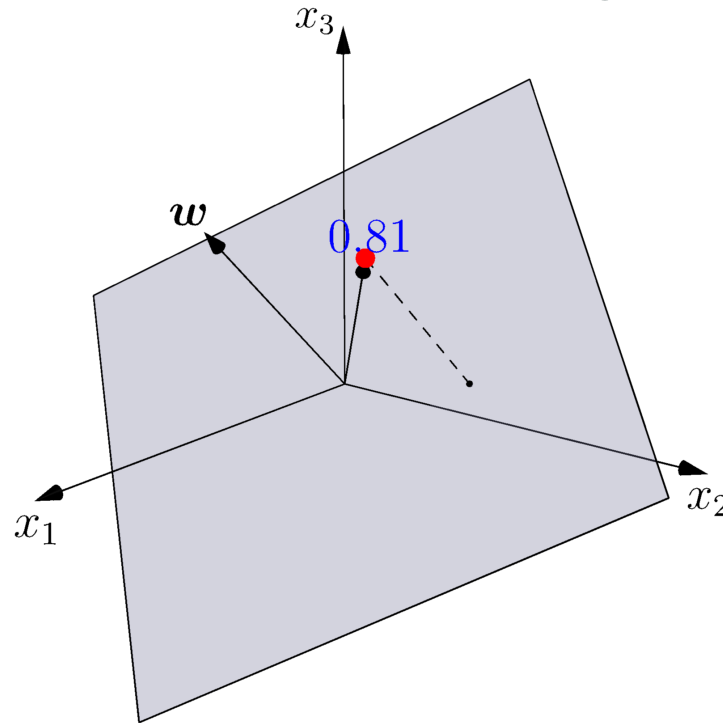
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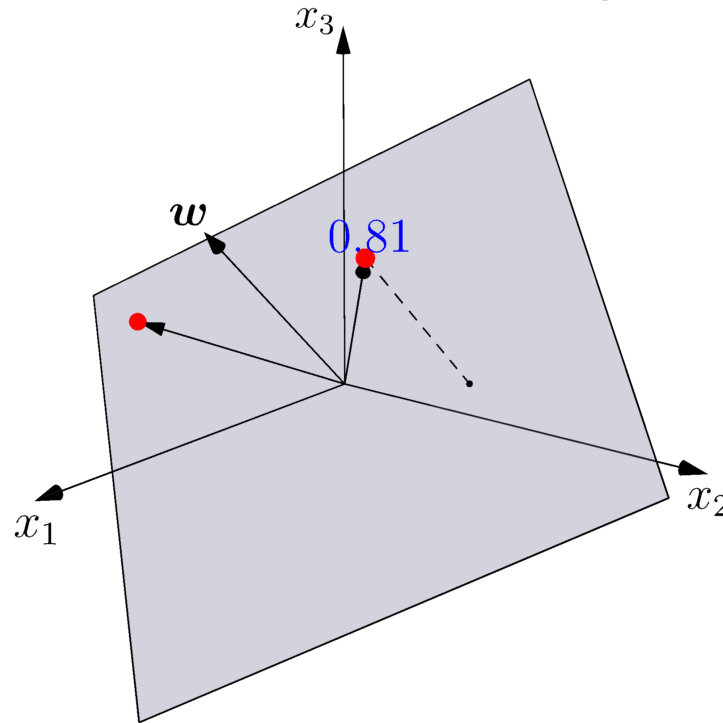
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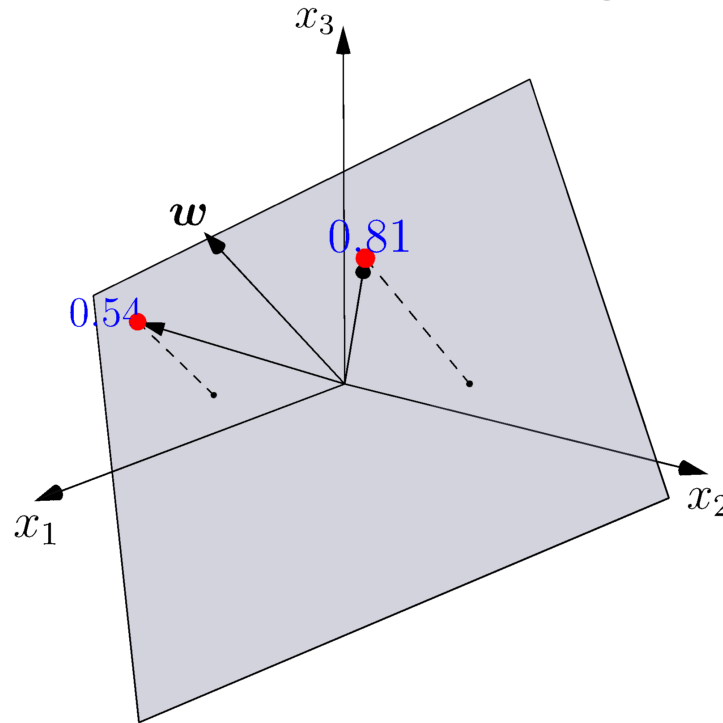
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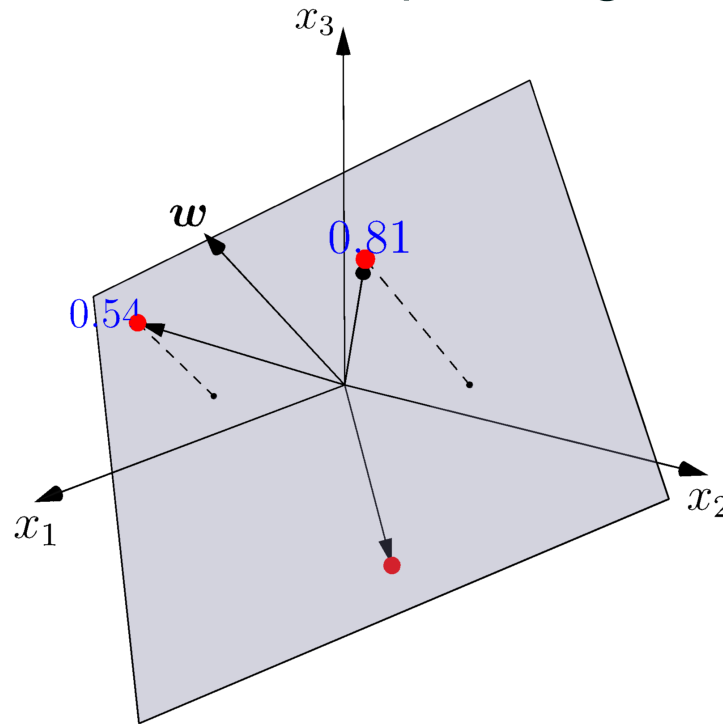
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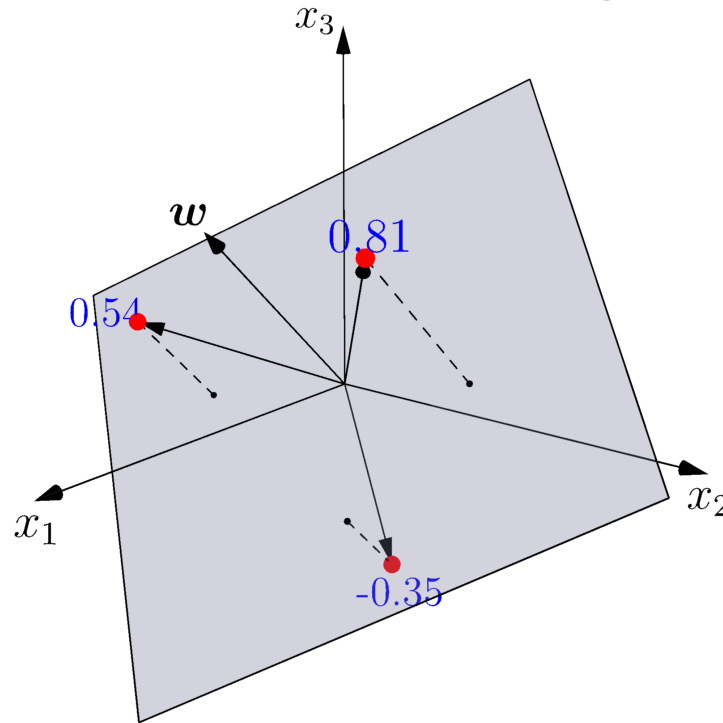
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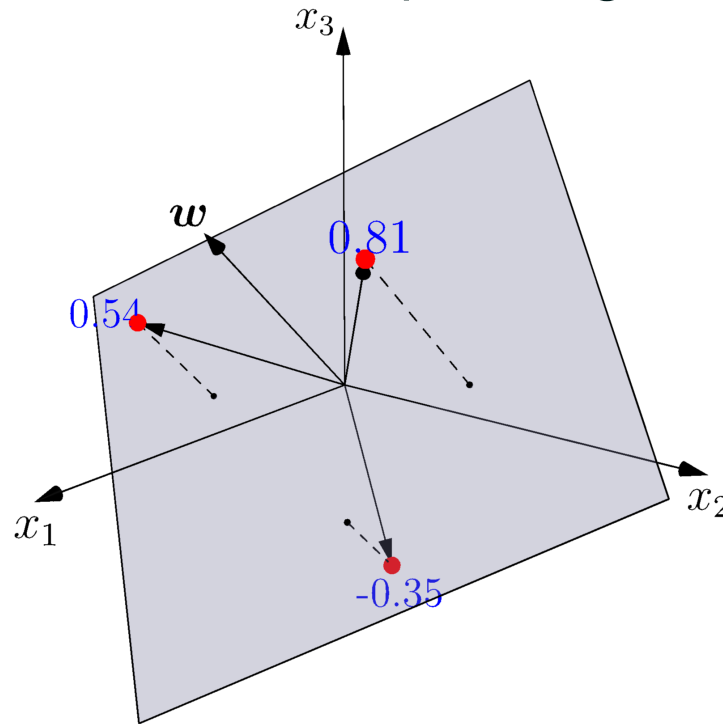
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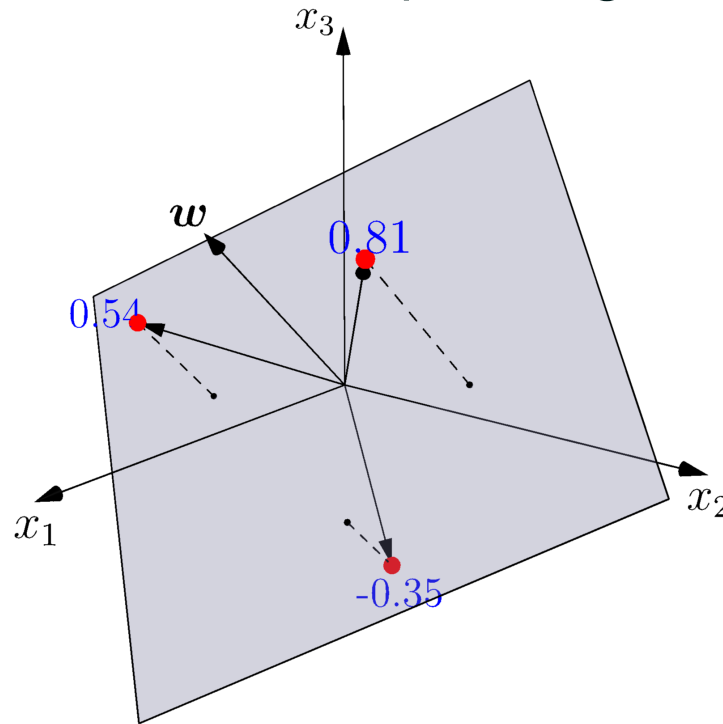
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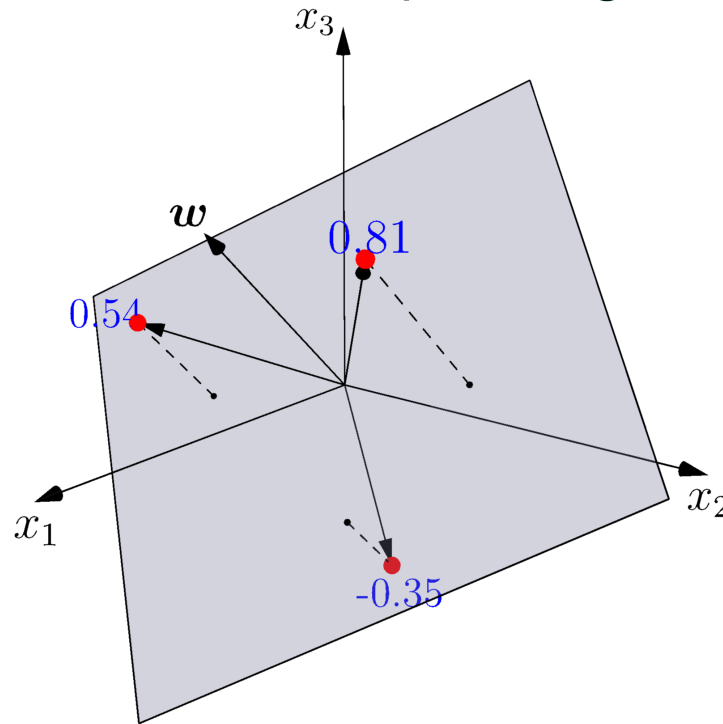
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# Linear Least Squares

- The error of input pattern  $\mathbf{x}_k$  is

$$\epsilon_k = \mathbf{x}_k^\top \mathbf{w} - y_k$$

- The squared error

$$E(\mathbf{w}|\mathcal{D}) = \sum_{k=1}^m (\mathbf{x}_k^\top \mathbf{w} - y_k)^2 = \sum_{k=1}^m \epsilon_k^2 = \|\boldsymbol{\epsilon}\|^2$$

- We can define the error vector

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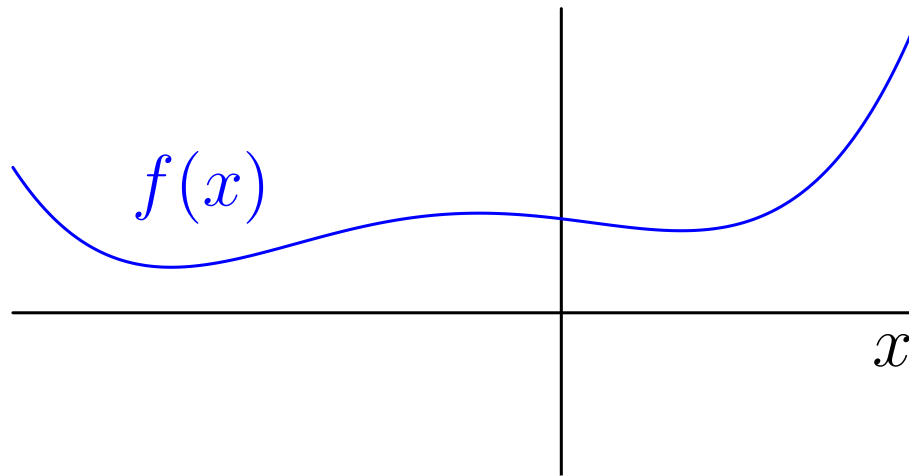
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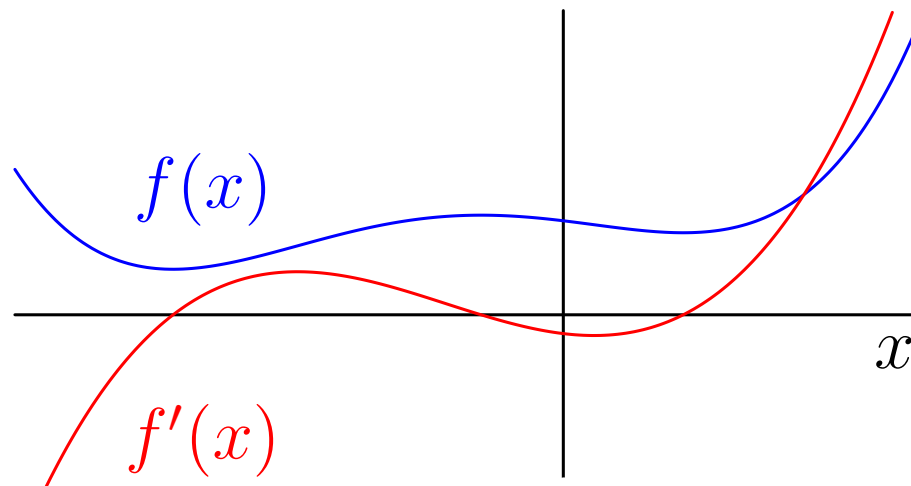
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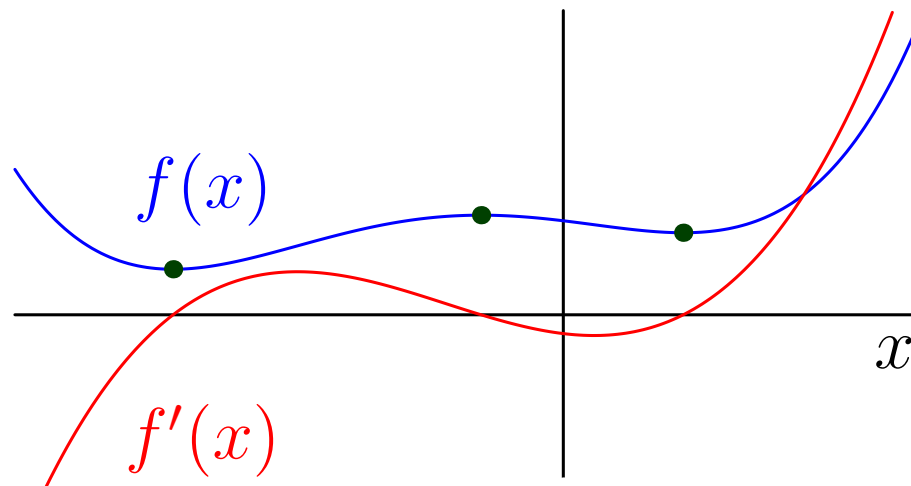
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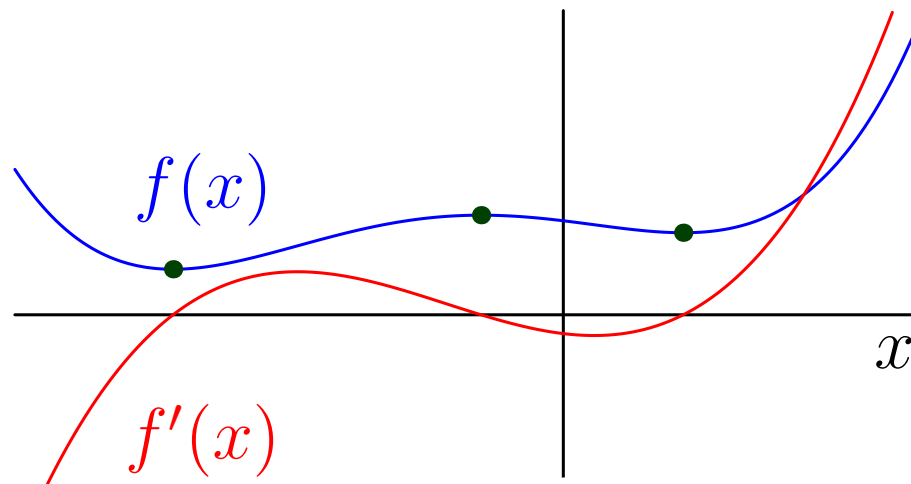
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- The minima of an  $n$ -dimensions function  $f(\mathbf{x})$  are given by the set of equations

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = 0 \quad \forall i = 1, \dots, n$$

# Gradients

- The **grad** operator  $\nabla$  is the gradient operator in high dimensions

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}$$

- The partial derivatives (curly d's)

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# Least Squares Solution

- The least squared solution is give by

$$\begin{aligned}\nabla E(\boldsymbol{w}|\mathcal{D}) &= \nabla \|\boldsymbol{\epsilon}\|^2 = \nabla \|\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}\|^2 \\ &= \nabla (\boldsymbol{w}^\top \boldsymbol{X}^\top \boldsymbol{X} \boldsymbol{w} - 2\boldsymbol{w}^\top \boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{y}) \\ &= 2(\boldsymbol{X}^\top \boldsymbol{X} \boldsymbol{w} - \boldsymbol{X}^\top \boldsymbol{y}) = 0\end{aligned}$$

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$$\boldsymbol{w} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{X}^+ \boldsymbol{y}$$

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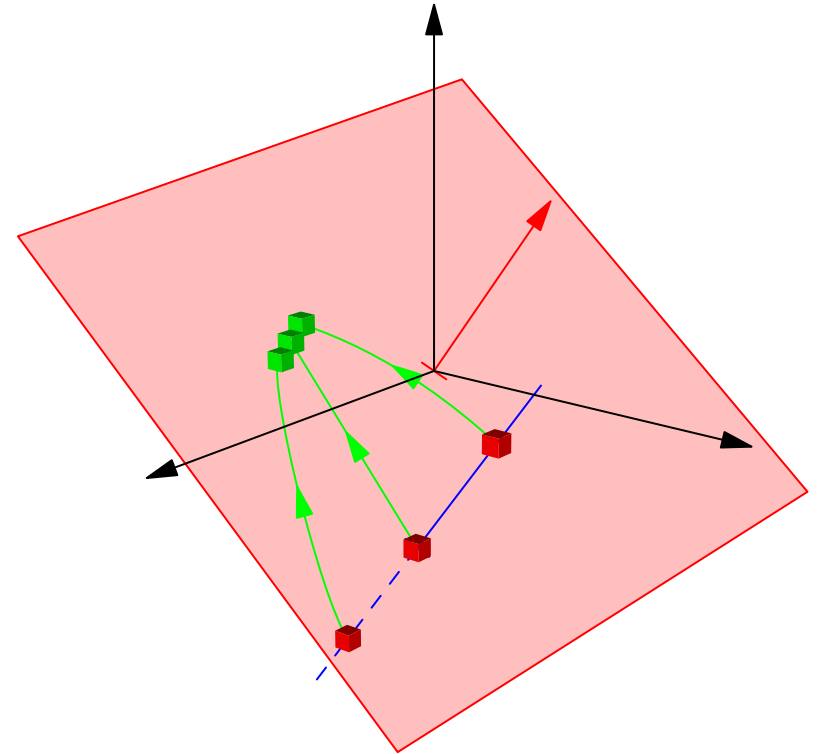
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- It is tedious to compute these things component-wise, but when you need to understand what is going on then go back to the basics

# Outline

1. Mappings
2. **Linear Maps**



# Solving Inverse Problems

- Gauss showed us how to solve **over-constrained** problems (we have more observations than parameters)
- We seek a solution which isn't necessarily exact but minimises an error
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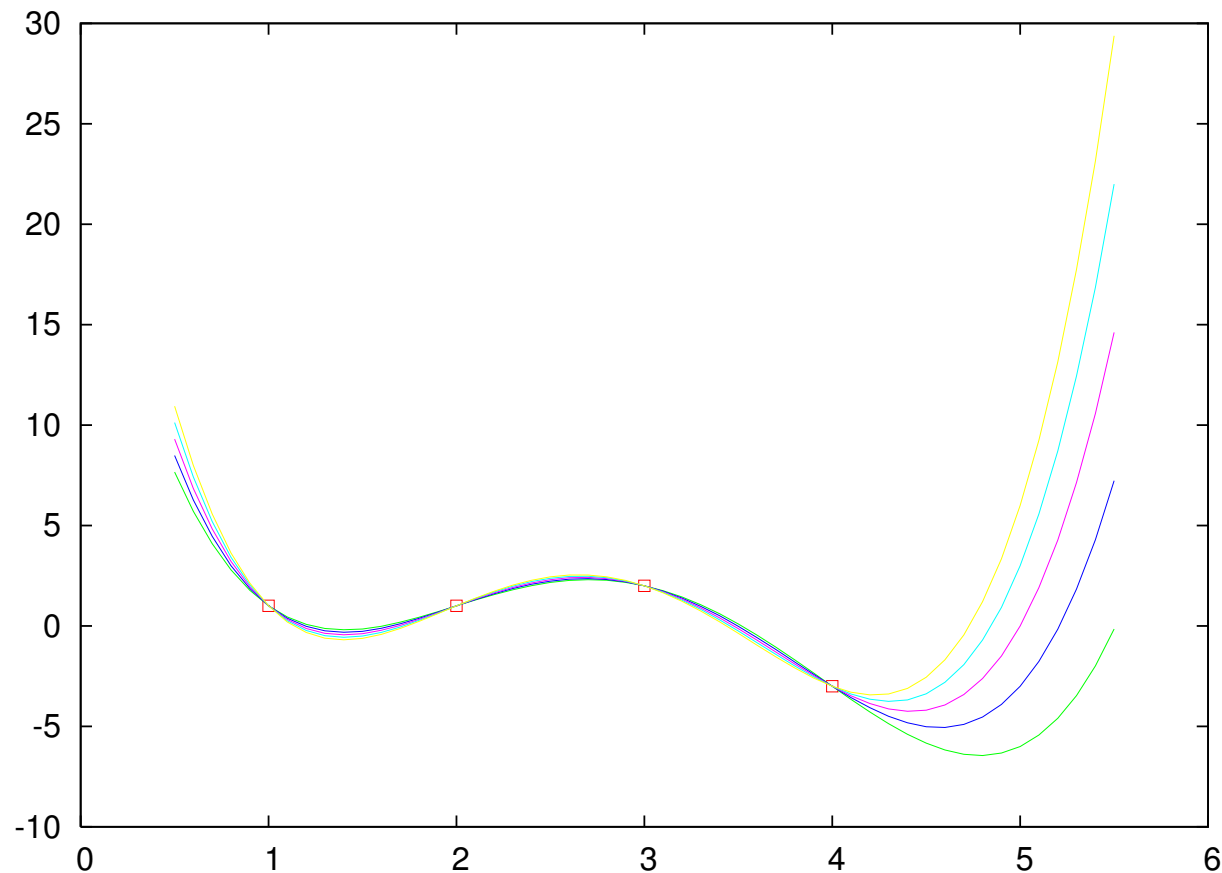
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- This is very typical of most machine learning problems

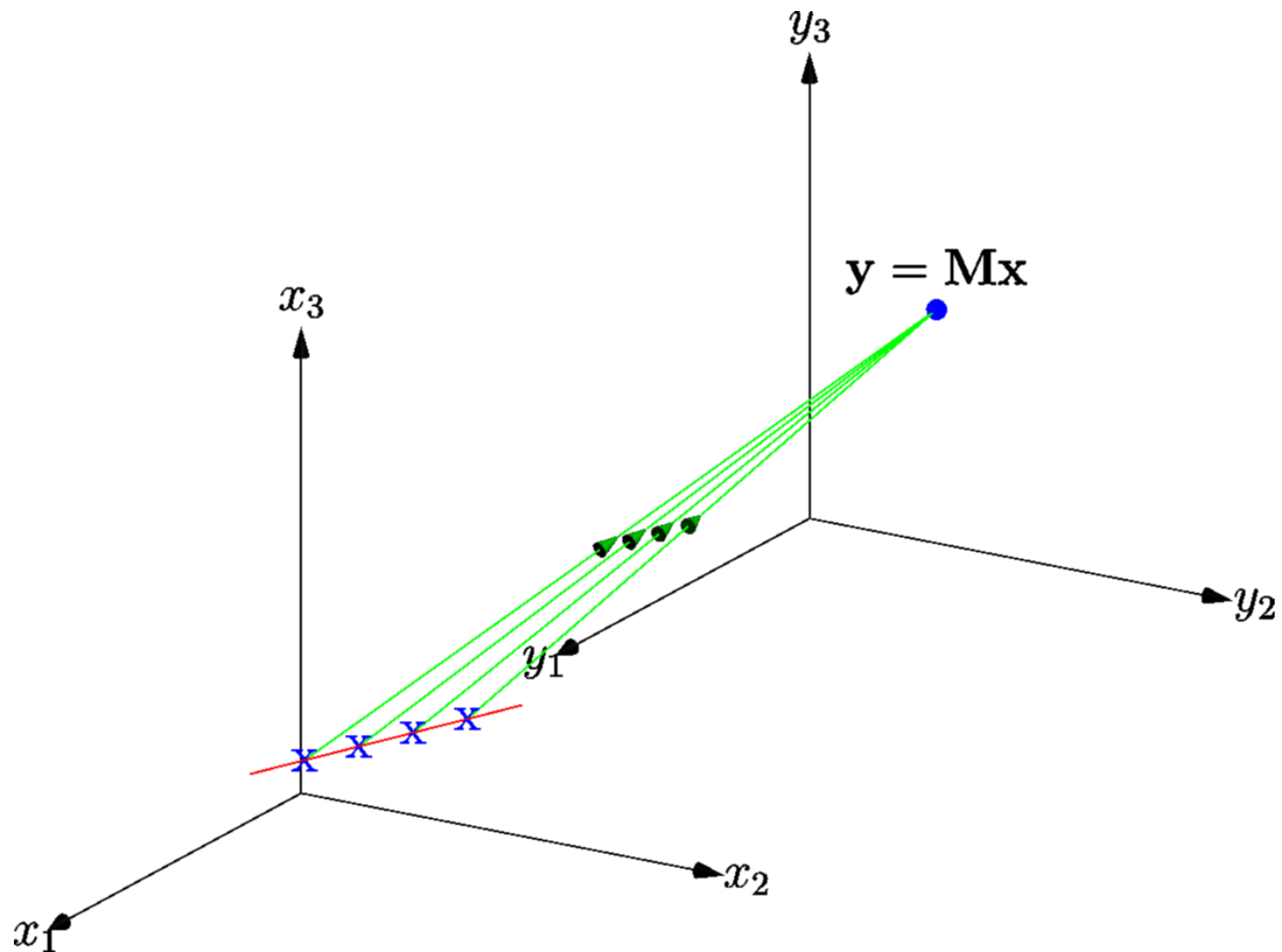
# Under Constrained Systems

- If we have less data-points than parameters then there will be multiple solutions



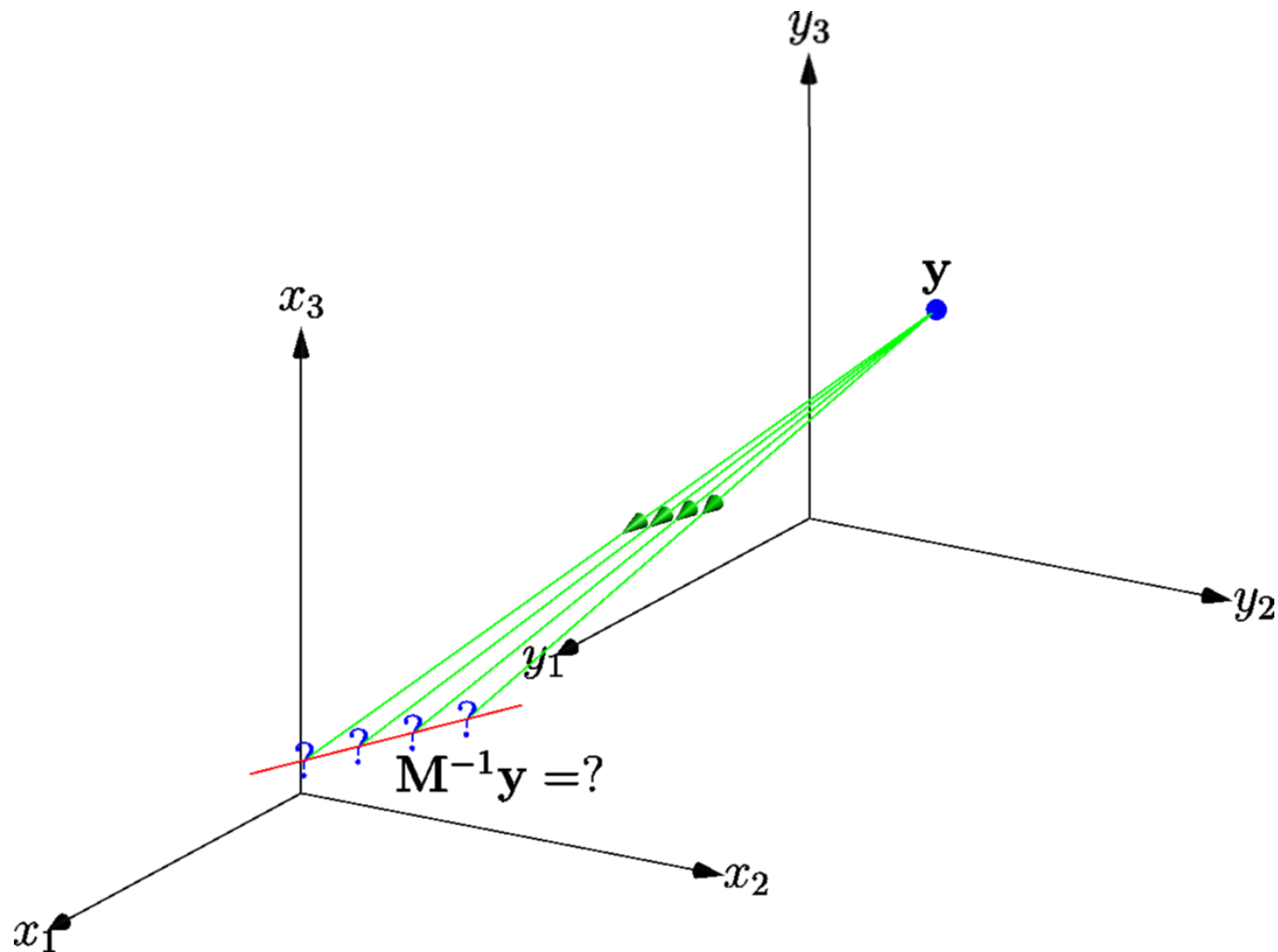
# What is the Inverse?

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# Under-constrained Systems

- The system is **under-constrained**
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- Solving the inverse problem ( $w = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ ) is said to be **ill-posed**
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- Singular matrices are rare (although they occur when we don't have enough data), but matrices that are close to being singular are common
- If a matrix is close to singular it is ill-conditioned
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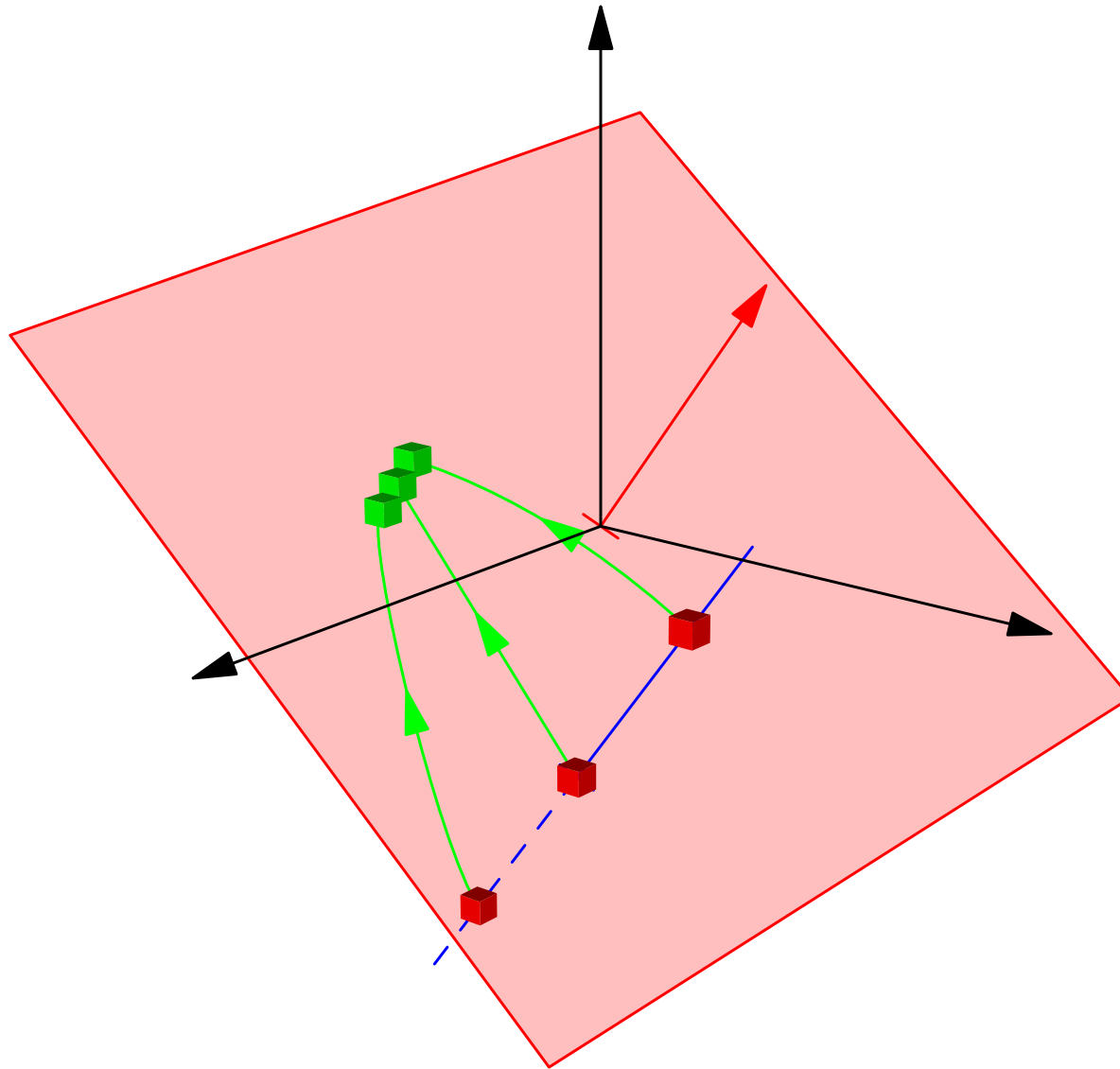
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# III-Conditioned Matrices



# III-Conditioning in ML

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- In linear regression the matrix  $\mathbf{X}^T\mathbf{X}$  is typically ill-conditioned when we have as many data points as parameters
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